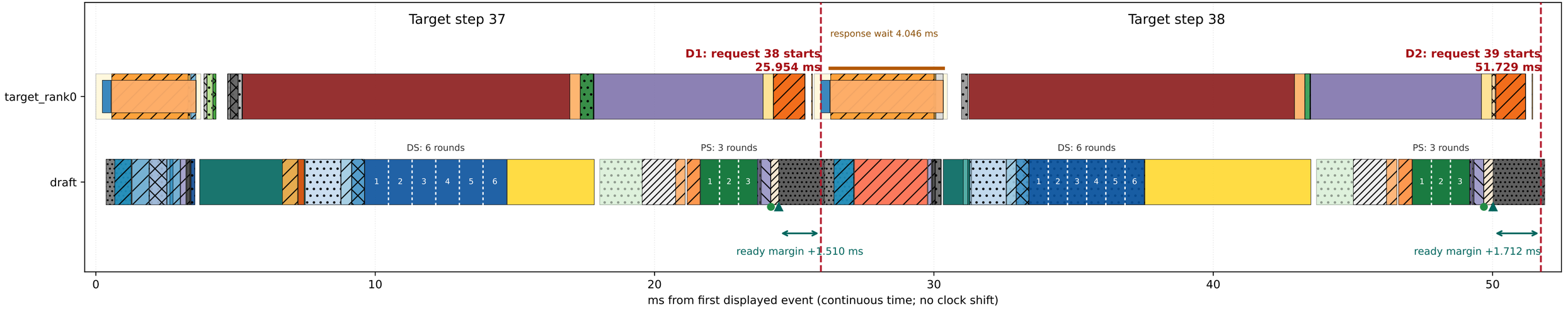
# LayerSkip Llama2-7B + llama-68M | P1 hit -> next step | UID 288_t0 | request 4

Historical v1.0 82d964e0 | exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.




| Prepared in step | Cache ready (ms) | Receive-ready (ms) | Next send (ms) | Cache margin (ms) | Ready margin (ms) | Next send span (ms) | Next response wait (ms) |
|---|---|---|---|---|---|---|---|
| 37 | 24.166 | 24.444 | 25.954 | +1.788 | +1.510 | 0.321 | 4.046 |
| 38 | 49.689 | 50.017 | 51.729 | +2.040 | +1.712 | 0.316 | 2.167 |

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

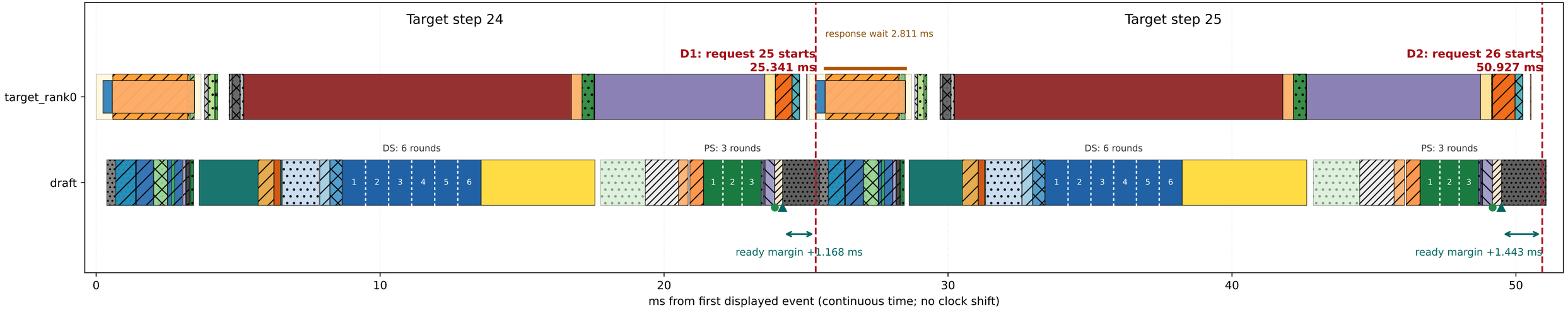
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

# LayerSkip Llama2-7B + llama-68M | P2 hit -> next step | UID 244_t0 | request 2

Historical v1.0 82d964e0 | exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

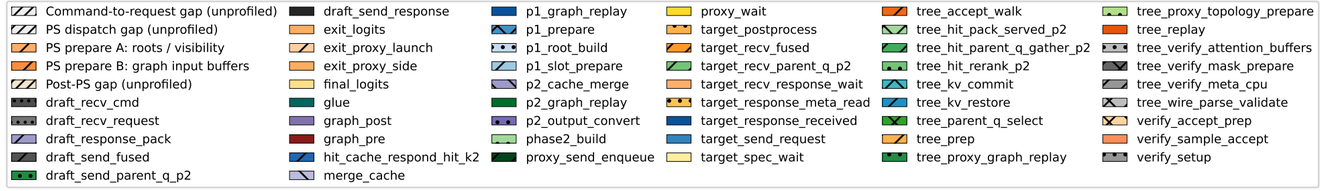
Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.




| Prepared in step | Cache ready (ms) | Receive-ready (ms) | Next send (ms) | Cache margin (ms) | Ready margin (ms) | Next send span (ms) | Next response wait (ms) |
|---|---|---|---|---|---|---|---|
| 24 | 23.903 | 24.173 | 25.341 | +1.438 | +1.168 | 0.323 | 2.811 |
| 25 | 49.180 | 49.484 | 50.927 | +1.746 | +1.443 | 0.327 | 2.818 |

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.




Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

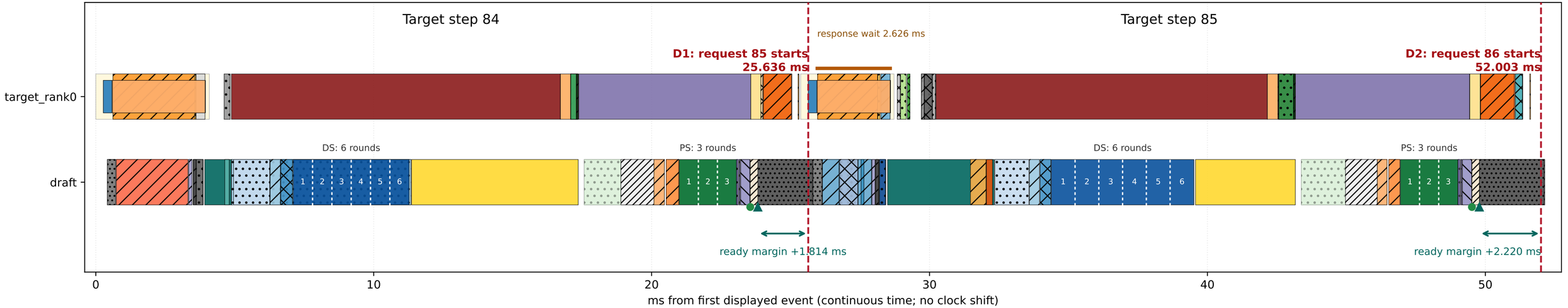
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

# LayerSkip Llama2-7B + llama-68M | Miss -> next step | UID 288_t0 | request 4

Historical v1.0 82d964e0 | exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.




| Prepared in step | Cache ready (ms) | Receive-ready (ms) | Next send (ms) | Cache margin (ms) | Ready margin (ms) | Next send span (ms) | Next response wait (ms) |
|---|---|---|---|---|---|---|---|
| 84 | 23.550 | 23.821 | 25.636 | +2.086 | +1.814 | 0.309 | 2.626 |
| 85 | 49.519 | 49.783 | 52.003 | +2.484 | +2.220 | 0.341 | 3.338 |

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_commit
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

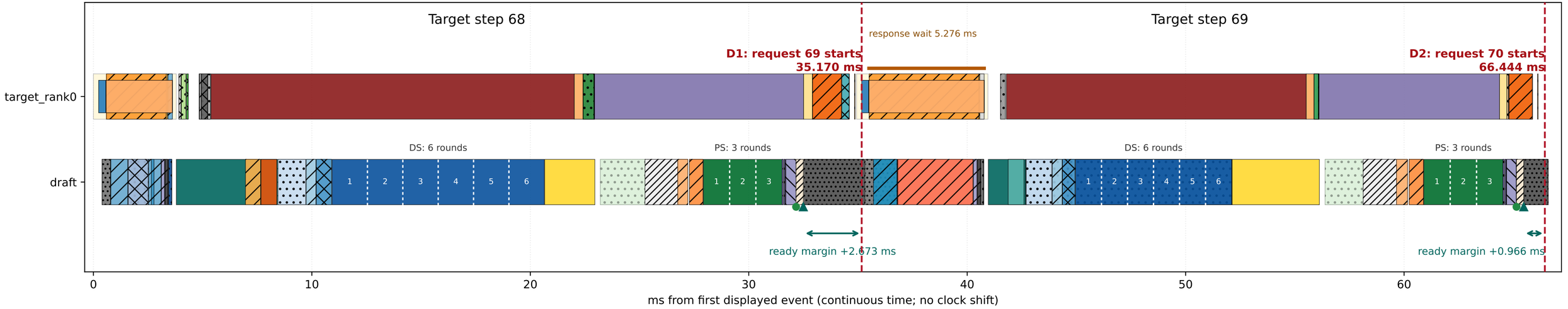
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

# LayerSkip Llama2-13B + AMD-Llama-135M | P1 hit -> next step | UID 84_t0 | request 1

Historical v1.0 59f46511 | exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

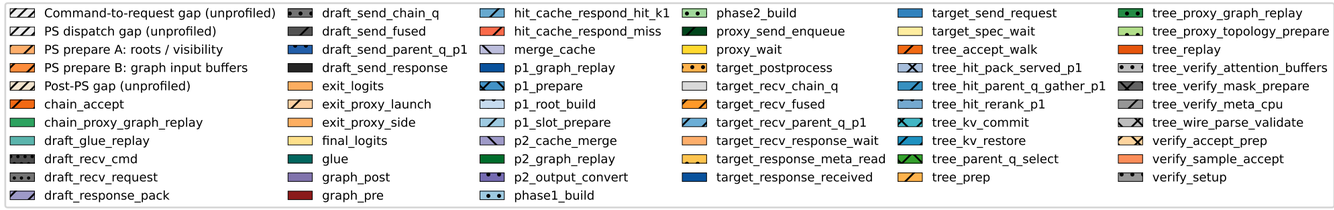
Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.




| Prepared in step | Cache ready (ms) | Receive-ready (ms) | Next send (ms) | Cache margin (ms) | Ready margin (ms) | Next send span (ms) | Next response wait (ms) |
|---|---|---|---|---|---|---|---|
| 68 | 32.167 | 32.497 | 35.170 | +3.003 | +2.673 | 0.314 | 5.276 |
| 69 | 65.147 | 65.478 | 66.444 | +1.297 | +0.966 | 0.321 | 2.856 |

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.




Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

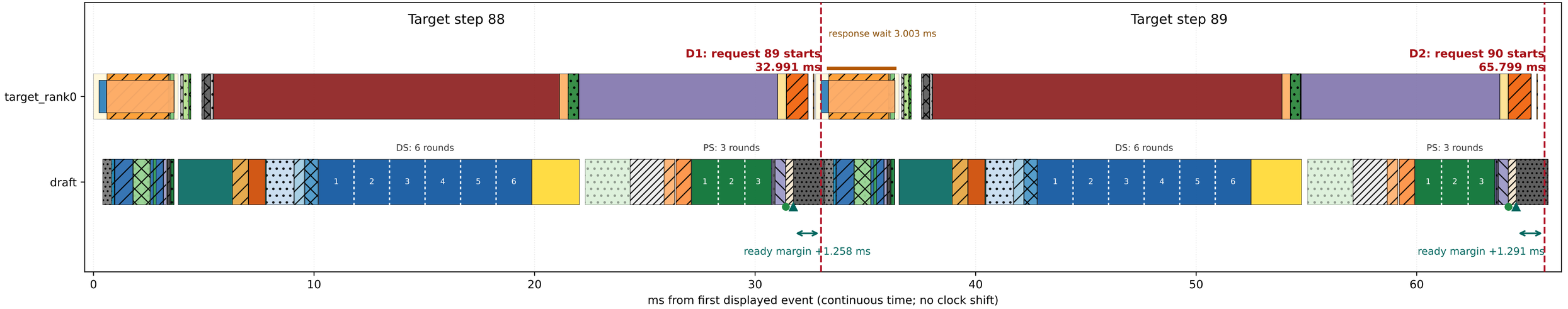
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

# LayerSkip Llama2-13B + AMD-Llama-135M | P2 hit -> next step | UID 244_t0 | request 2

Historical v1.0 59f46511 | exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.




| Prepared in step | Cache ready (ms) | Receive-ready (ms) | Next send (ms) | Cache margin (ms) | Ready margin (ms) | Next send span (ms) | Next response wait (ms) |
|---|---|---|---|---|---|---|---|
| 88 | 31.394 | 31.733 | 32.991 | +1.597 | +1.258 | 0.322 | 3.003 |
| 89 | 64.164 | 64.508 | 65.799 | +1.635 | +1.291 | 0.346 | 2.953 |

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

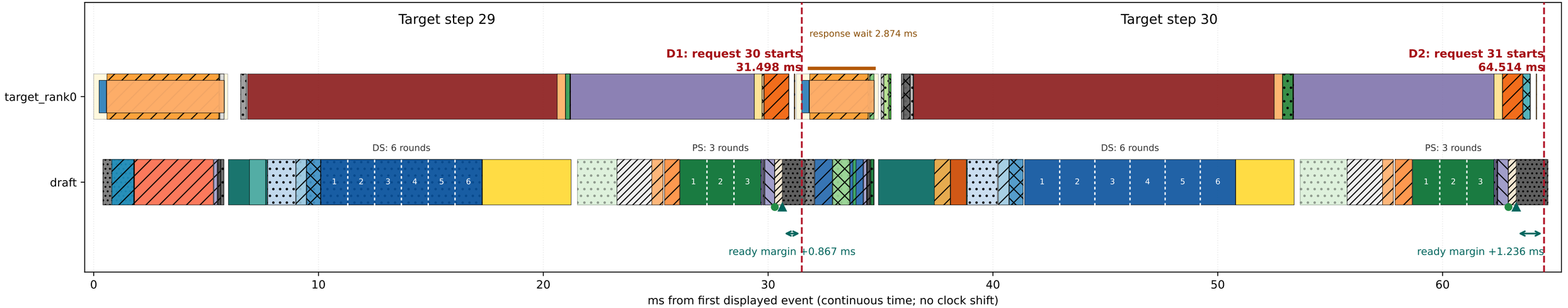
Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

# LayerSkip Llama2-13B + AMD-Llama-135M | Miss -> next step | UID 128_t0 | request 3

Historical v1.0 59f46511 | exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.




| Prepared in step | Cache ready (ms) | Receive-ready (ms) | Next send (ms) | Cache margin (ms) | Ready margin (ms) | Next send span (ms) | Next response wait (ms) |
|---|---|---|---|---|---|---|---|
| 29 | 30.298 | 30.631 | 31.498 | +1.200 | +0.867 | 0.322 | 2.874 |
| 30 | 62.933 | 63.277 | 64.514 | +1.581 | +1.236 | 0.347 | 3.728 |

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

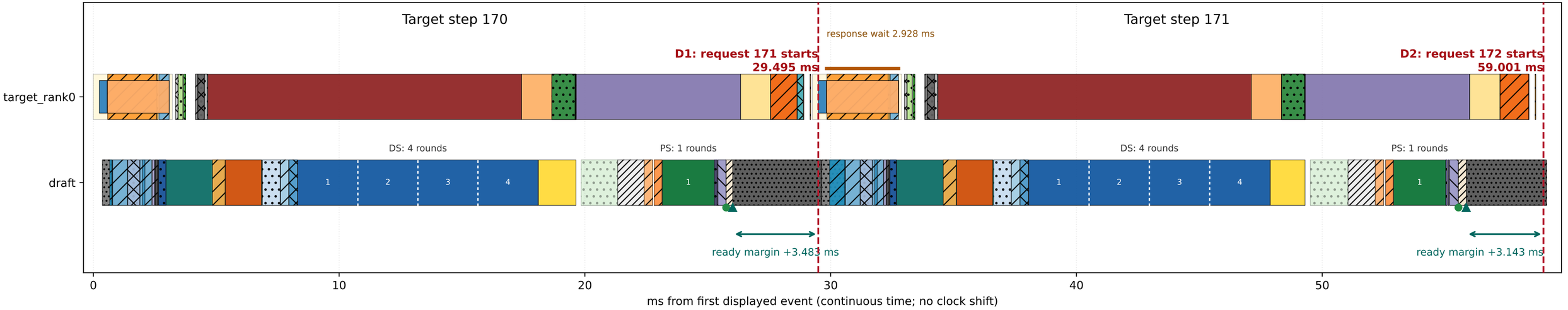
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

# LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M | P1 hit -> next step | UID 84_t0 | request 1

Historical v1.0 16dd4198 | exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

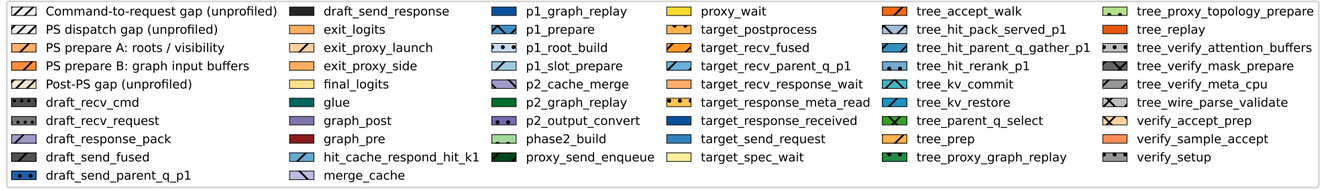
Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.




| Prepared in step | Cache ready (ms) | Receive-ready (ms) | Next send (ms) | Cache margin (ms) | Ready margin (ms) | Next send span (ms) | Next response wait (ms) |
|---|---|---|---|---|---|---|---|
| 170 | 25.749 | 26.012 | 29.495 | +3.746 | +3.483 | 0.323 | 2.928 |
| 171 | 55.542 | 55.858 | 59.001 | +3.459 | +3.143 | 0.337 | 2.215 |

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.




Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

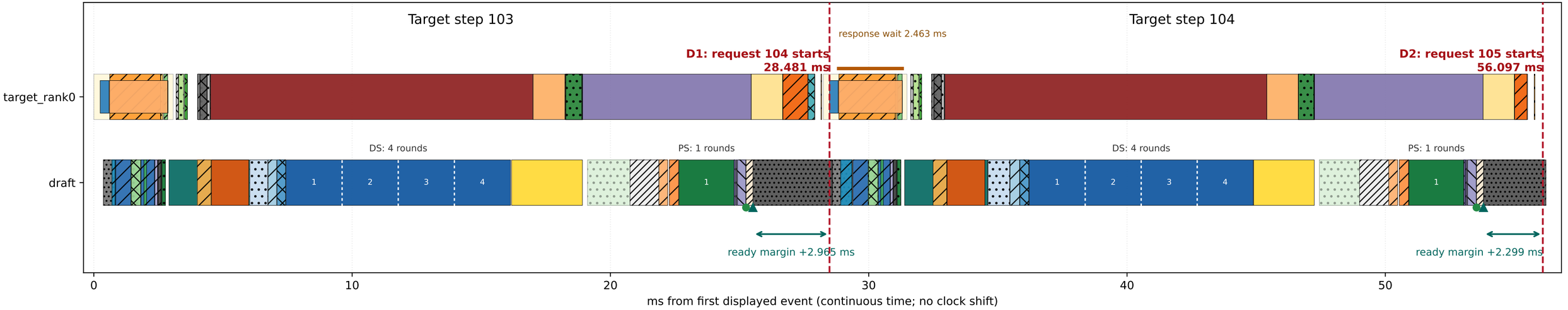
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

# LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M | P2 hit -> next step | UID 84_t0 | request 1

Historical v1.0 16dd4198 | exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

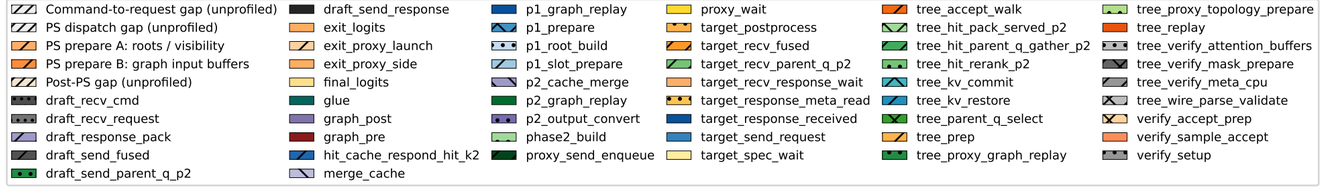
Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.




| Prepared in step | Cache ready (ms) | Receive-ready (ms) | Next send (ms) | Cache margin (ms) | Ready margin (ms) | Next send span (ms) | Next response wait (ms) |
|---|---|---|---|---|---|---|---|
| 103 | 25.251 | 25.517 | 28.481 | +3.230 | +2.965 | 0.336 | 2.463 |
| 104 | 53.531 | 53.798 | 56.097 | +2.566 | +2.299 | 0.319 | 2.479 |

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.




Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

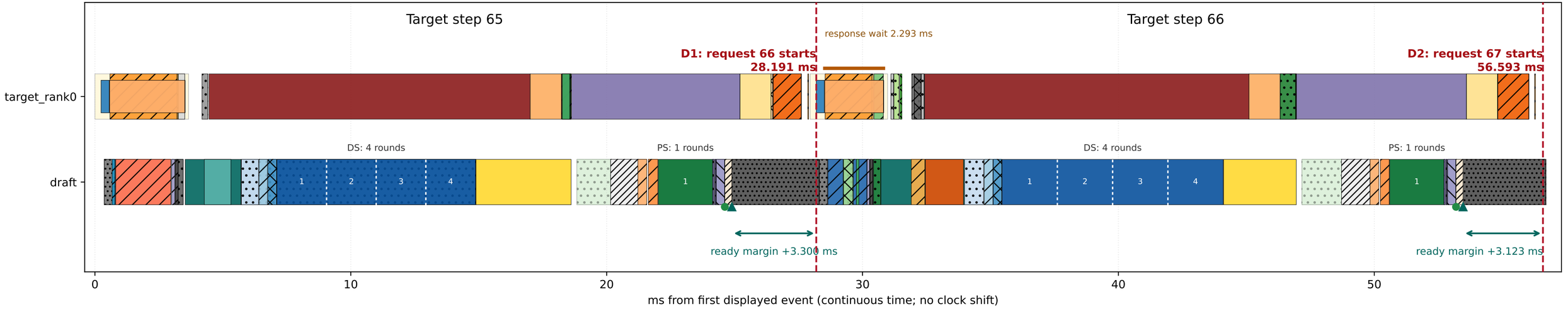
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

# LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M | Miss -> next step | UID 244_t0 | request 2

Historical v1.0 16dd4198 | exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.




| Prepared in step | Cache ready (ms) | Receive-ready (ms) | Next send (ms) | Cache margin (ms) | Ready margin (ms) | Next send span (ms) | Next response wait (ms) |
|---|---|---|---|---|---|---|---|
| 65 | 24.619 | 24.892 | 28.191 | +3.572 | +3.300 | 0.315 | 2.293 |
| 66 | 53.196 | 53.470 | 56.593 | +3.397 | +3.123 | 0.316 | 2.146 |

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

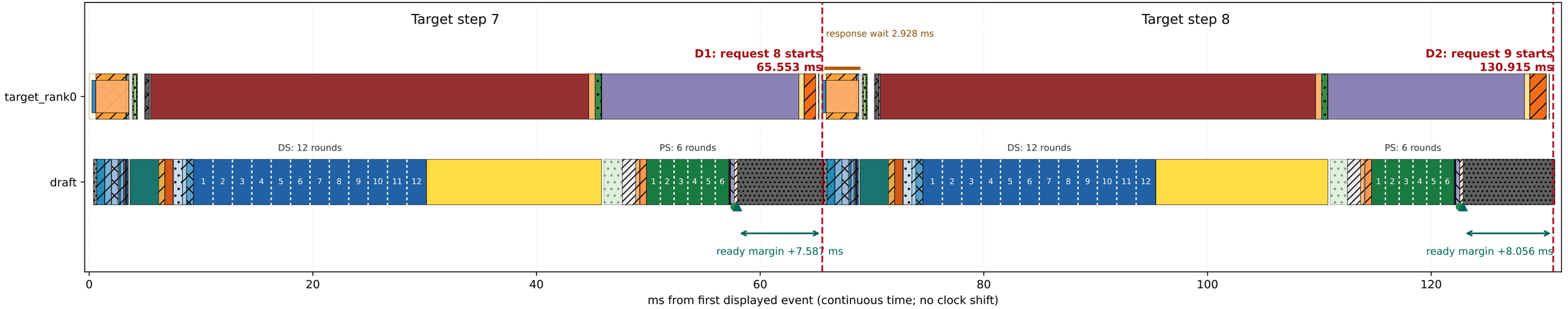
Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

# LayerSkip CodeLlama-34B + AMD-Llama-135M-code | P1 hit -> next step | UID humaneval_local_033_t0 | request 4

Historical v1.0 59f46511 | exit 30 (0-based) | K1/K2 12/6 | G=M DS/PS 12/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.




| Prepared in step | Cache ready (ms) | Receive-ready (ms) | Next send (ms) | Cache margin (ms) | Ready margin (ms) | Next send span (ms) | Next response wait (ms) |
|---|---|---|---|---|---|---|---|
| 7 | 57.692 | 57.966 | 65.553 | +7.860 | +7.587 | 0.319 | 2.928 |
| 8 | 122.555 | 122.859 | 130.915 | +8.360 | +8.056 | 0.325 | 3.104 |

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

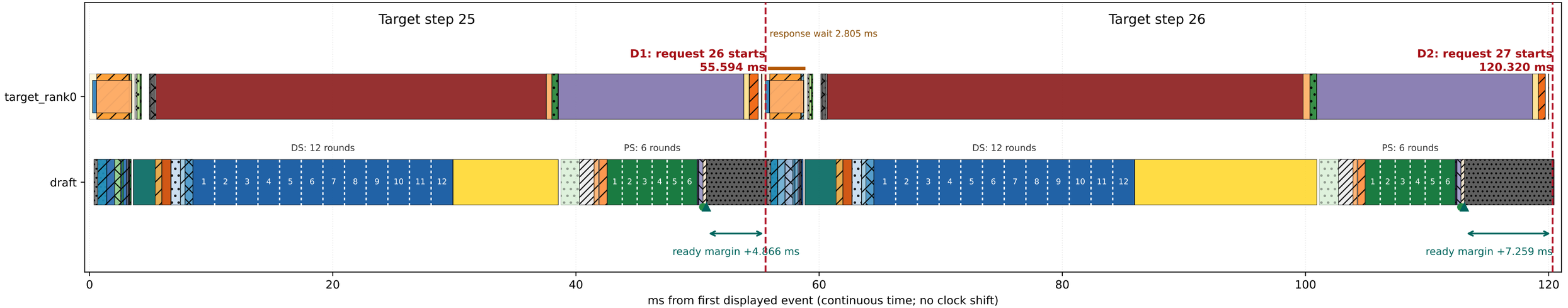
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

# LayerSkip CodeLlama-34B + AMD-Llama-135M-code | P2 hit -> next step | UID humaneval_local_023_t0 | request 3

Historical v1.0 59f46511 | exit 30 (0-based) | K1/K2 12/6 | G=M DS/PS 12/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.




| Prepared in step | Cache ready (ms) | Receive-ready (ms) | Next send (ms) | Cache margin (ms) | Ready margin (ms) | Next send span (ms) | Next response wait (ms) |
|---|---|---|---|---|---|---|---|
| 25 | 50.453 | 50.728 | 55.594 | +5.141 | +4.866 | 0.318 | 2.805 |
| 26 | 112.792 | 113.062 | 120.320 | +7.528 | +7.259 | 0.321 | 2.240 |

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p1
tree_hit_parent_q_gather_p2
tree_hit_rerank_p1
tree_hit_rerank_p2
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

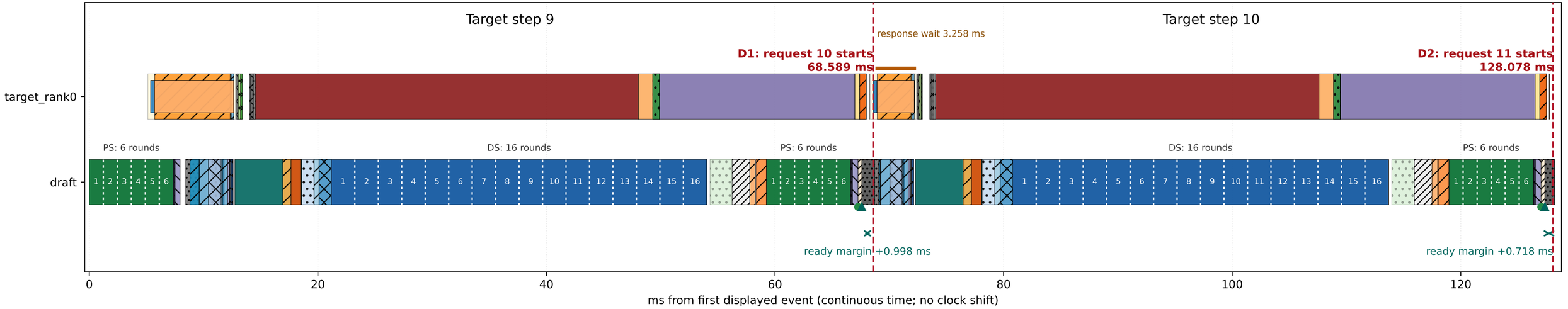
Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

# CodeLlama-34B + AMD-135M-code / throughput alternative | P1 hit -> next step | UID humaneval_local_023_t0 | request 3

Historical v1.0 59f46511 | exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

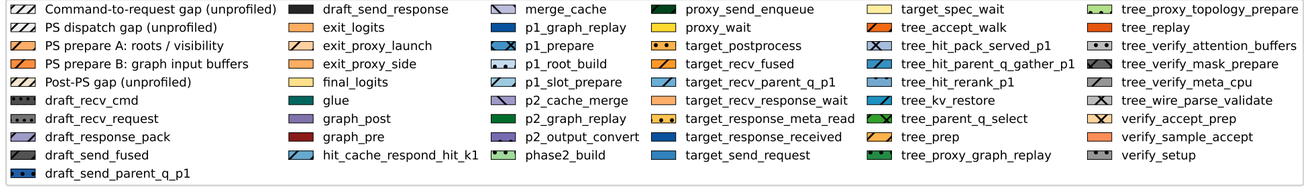
Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.




| Prepared in step | Cache ready (ms) | Receive-ready (ms) | Next send (ms) | Cache margin (ms) | Ready margin (ms) | Next send span (ms) | Next response wait (ms) |
|---|---|---|---|---|---|---|---|
| 9 | 67.265 | 67.590 | 68.589 | +1.324 | +0.998 | 0.329 | 3.258 |
| 10 | 127.035 | 127.359 | 128.078 | +1.043 | +0.718 | 0.326 | 3.228 |

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.




Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

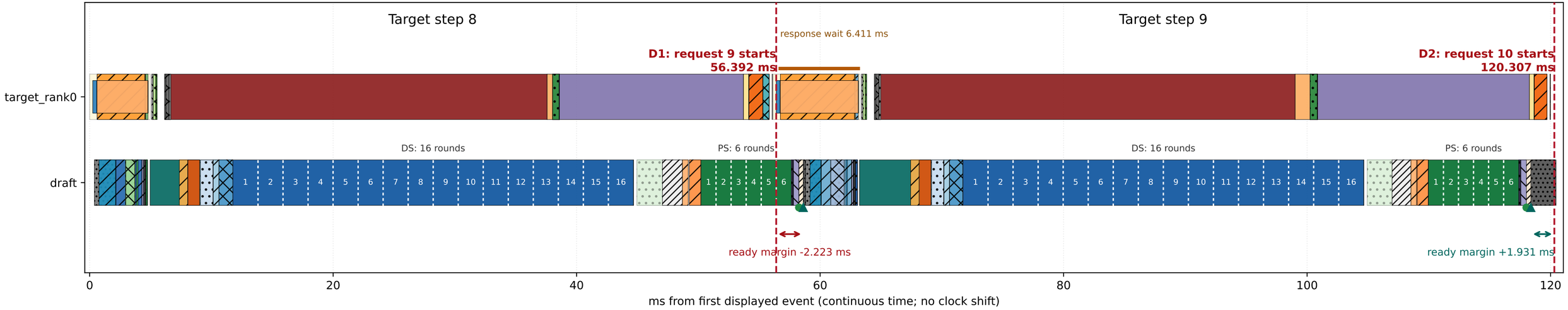
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

# CodeLlama-34B + AMD-135M-code / throughput alternative | P2 hit -> next step | UID humaneval_local_003_t0 | request 1

Historical v1.0 59f46511 | exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.




| Prepared in step | Cache ready (ms) | Receive-ready (ms) | Next send (ms) | Cache margin (ms) | Ready margin (ms) | Next send span (ms) | Next response wait (ms) |
|---|---|---|---|---|---|---|---|
| 8 | 58.265 | 58.615 | 56.392 | -1.873 | -2.223 | 0.315 | 6.411 |
| 9 | 118.013 | 118.376 | 120.307 | +2.295 | +1.931 | 0.328 | 4.405 |

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p1
tree_hit_parent_q_gather_p2
tree_hit_rerank_p1
tree_hit_rerank_p2
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

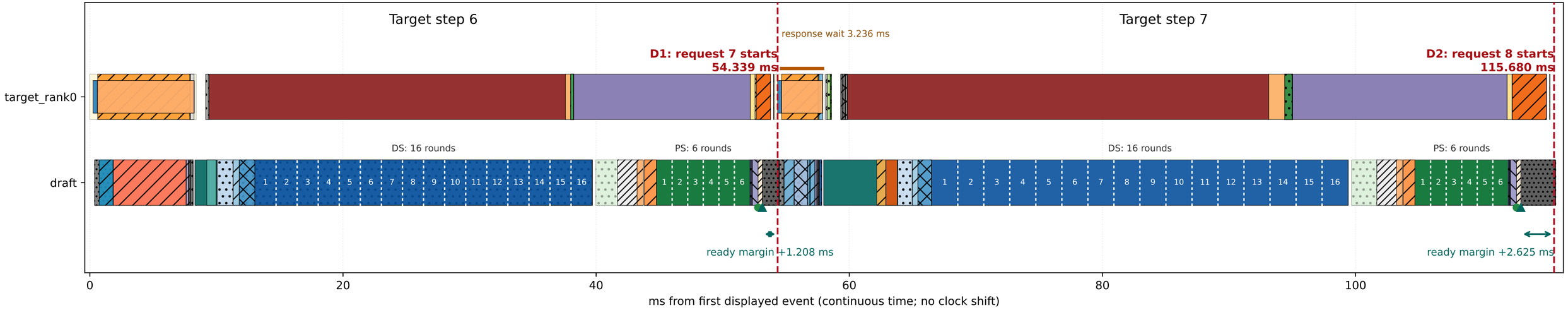
Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.
Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

# CodeLlama-34B + AMD-135M-code / throughput alternative | Miss -> next step | UID humaneval_local_023_t0 | request 3

Historical v1.0 59f46511 | exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.




| Prepared in step | Cache ready (ms) | Receive-ready (ms) | Next send (ms) | Cache margin (ms) | Ready margin (ms) | Next send span (ms) | Next response wait (ms) |
|---|---|---|---|---|---|---|---|
| 6 | 52.795 | 53.131 | 54.339 | +1.544 | +1.208 | 0.296 | 3.236 |
| 7 | 112.719 | 113.055 | 115.680 | +2.961 | +2.625 | 0.337 | 4.271 |

Margin = next send start minus readiness. Positive means ready before the observed request; negative means preparation ran past it.

This is not a causal stall estimate or a promise that all margin can be used for more forwards. The second next-response wait is outside the two-step view.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.